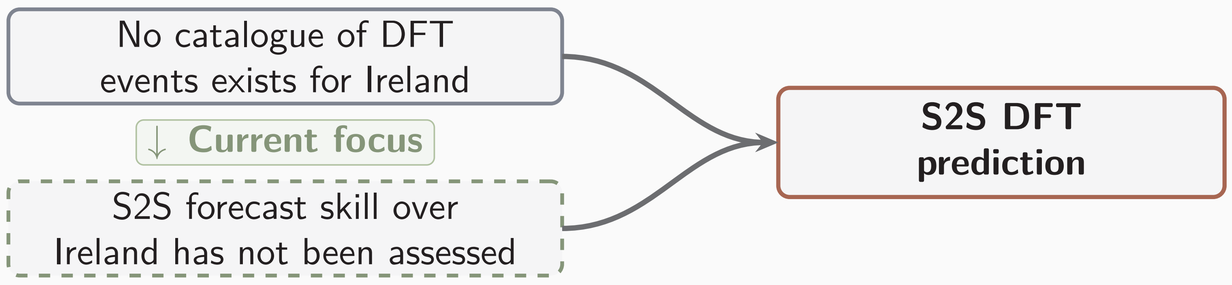
Towards Subseasonal-to-Seasonal Drought-to-Flood Prediction

Lainey Ward[1], Fiachra O'Loughlin[1], Conor Sweeney[2]

[1]School of Civil Engineering, [2]School of Mathematics and Statistics, University College Dublin

Introduction

Floods and droughts can occur in sequence — a **drought-to-flood transition (DFT)**. Globally, **~25% of floods are preceded by drought** [2], and socio-economic losses can be up to **8x greater** than from isolated hazards [3]. Forecasting DFTs at subseasonal-to-seasonal (S2S) lead times could support early action, but 2 research gaps remain ⟶



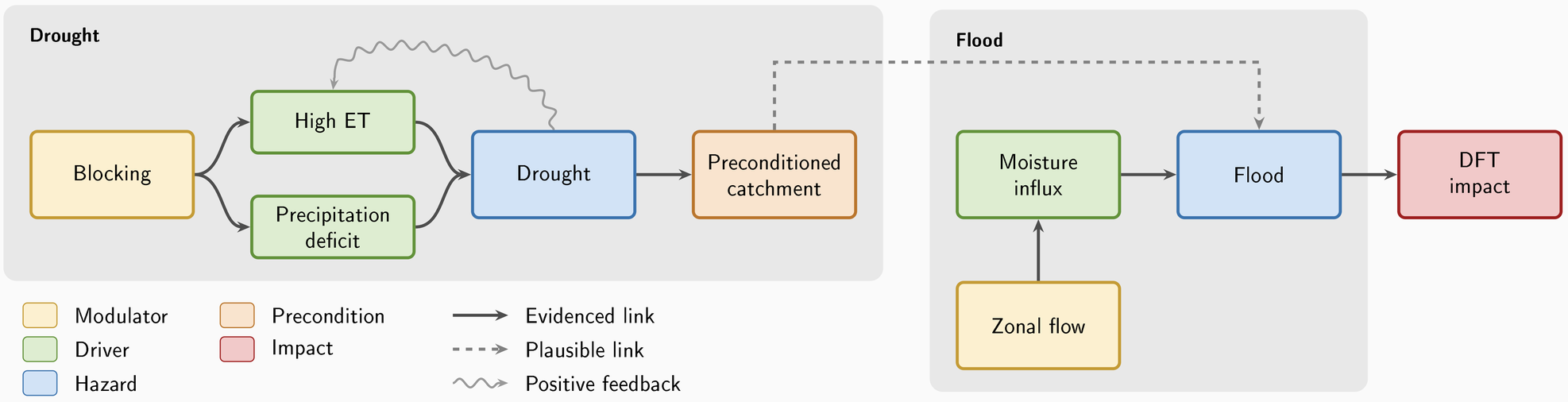


Figure 1: Physical pathway for DFT events, after the framework of Bevacqua et al. [1].

Methods

We verify 2 m temperature forecasts at **2–6 week** lead times, pooling all reforecast initialisations (2005–2024) per season and lead. Two complementary metrics are used: **Fair CRPSS** (Continuous Ranked Probability Skill Score) [4], scored against ERA5 climatology, and **centred ACC** (Anomaly Correlation Coefficient). Mean bias is removed by leave-one-year-out correction.

$$\mathrm{ACC} = \frac{\sum_{m=1}^{M} w_m (f'_m - \bar{f}')(o'_m - \bar{o}')}{\left[\sum_{m=1}^{M} w_m (f'_m - \bar{f}')^2 \cdot \sum_{m=1}^{M} w_m (o'_m - \bar{o}')^2\right]^{1/2}}$$

f'_m, o'_m = forecast/observed anomalies
$\bar{f}'$, $\bar{o}'$ = area-weighted spatial means
$w_m = \cos\varphi$ (latitude weight)
M = number of gridpoints

Wilks (2011), Eq. 7.59 [5]

Data

System	ECMWF ENS extended-range
IFS cycle	CY49R1 (2024)
Init. freq.	Every 2 days
Members	11 (1 control + 10 perturbed)
Horizon	Weeks 2–6
Variable	2 m temperature
Grid	0.28° regular lat-lon (MARS)
Reference	ERA5 + Met Éireann stations
Reforecasts	2005–2024

Results

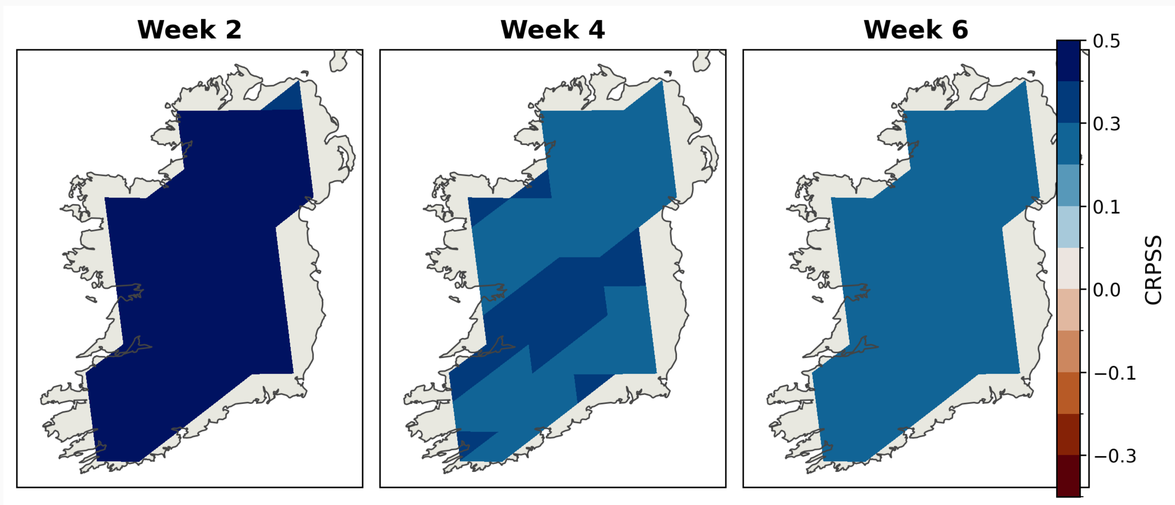


Figure 2: Fair CRPSS for ENS extended 2 m temperature at Weeks 2, 4, and 6. Skill decays with lead time but remains positive across Ireland through Week 6.

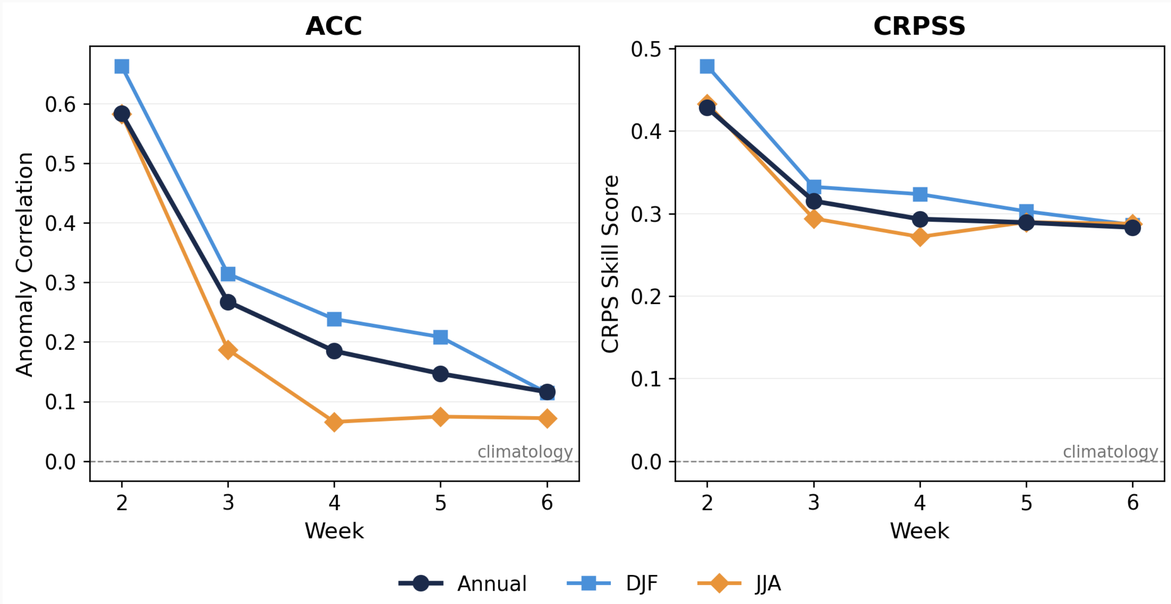


Figure 3: Spatial ACC and domain-mean CRPSS vs lead time. Dashed line marks ACC = 0.6 ("useful skill"; Hollingsworth et al. [6]). DJF shows highest deterministic skill; probabilistic skill positive through Week 6.

Discussion & Future Work

- First verification of S2S forecast skill over Ireland
- DJF strongest deterministic skill; JJA fades by Week 4
- Probabilistic skill robust through Week 6 across all seasons

- Compare with SEAS5 seasonal forecasts to map skill across the full S2S timescale
- Create DFT event catalogue; link to catchment and soil properties

⟶ **Identify predictable at-risk catchments**

References

[1] Bevacqua et al. (2021). *Earth's Future*, 9, e2021EF002340, [2] Matanó et al. (2024). *Environ. Res. Lett.*, 19, 064048, [3] Worou & Messori (2025). *Environ. Res. Lett.*, 20, 104024, [4] Ferro (2014). *Q. J. R. Meteorol. Soc.*, 140, 1917–1923, [5] Wilks (2011). *Statistical Methods in the Atmospheric Sciences*, 3rd ed., Eq. 7.59, [6] Hollingsworth et al. (1980). *Mon. Wea. Rev.*, 108, 1736–1773.